



Republic of the Philippines
Pamantasan ng Cabuyao
(UNIVERSITY OF CABUYAO)
Academic Affairs Division
Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

OUTCOMES BASED TEACHING AND LEARNING PLAN
in Programming Logic and Design
Department: Computer Engineering

VISION	CORE VALUES
An institution of higher learning in Region IV, developing globally-competitive and value-laden professionals and leaders instrumental to community development and nation building.	As a God-fearing institution respecting multi-faith of people, PnC adheres to the following core values: • Personal Dignity • Nurturing Community • Commitment to Excellence
MISSION	PnC (UC) GRADUATE ATTRIBUTES
An institution of higher learning committed to equip individuals with knowledge, skills and values that will enable them to achieve professional goals & provide leadership and service for national development.	1. Professional competent individual 2. Proficient communicator 3. Analytical & problem solver 4. Lifelong learner 5. Team leader or player 6. Productive individual 7. Socially and ethically responsible citizen 8. Technology-driven professional 9. Value-laden professional



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Program Educational Objectives	<i>Three to five years after graduation, the Bachelor of Science in Computer Engineering alumni shall::</i> <i>1. be employed in the field of artificial intelligence or embedded engineering or software engineering;</i> <i>2. possess globally competitive skills;</i> <i>3. demonstrate personal dignity and integrity; and</i> <i>4. engage in community deviation-building.</i>				
Course Intended Learning Outcomes (CILO)	<i>1. Identify important steps in program development cycle.</i> <i>2. Draw a flowchart to represent the program's logic.</i> <i>3. Break down programming problems into modules.</i>				
Student Outcomes Map Addressed by the Course	SO 3.	Design solutions for complex engineering problems and design systems, components, or processes to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, cultural, and sustainability, in accordance with standards	I/L	E/P	D/O
			I		

Legend:
I/L – Introductory/Learning
E/P – Enabling/Practice
D/O – Demonstrating/Opportunity



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COURSE/SUBJECT CODE	CPP102	COURSE/SUBJECT TITLE	PROGRAMMING LOGIC AND DESIGN
CREDIT UNITS	2	COURSE/SUBJECT PRE-REQUISITE/CO-REQUISITE	NONE
COURSE/SUBJECT DESCRIPTION	This is an introductory course in computer programming logic. The student will learn algorithms applicable to all programming languages, including identifiers, data types, arrays, control structures, modular programming, generating reports, and computer memory concepts. The student will learn to use charts commonly used in business and information processing. Program logic will be developed using flowcharts and pseudo code. Programs will be written using any programming language.		

TEACHING AND LEARNING PLAN

Week	Topics	Intended Learning Outcomes (ILO)	Teaching and Learning Activities (TLA)	Instructional Material	Assessment Tasks (AT)
1	Class orientation Introduction to Programming Language	1. Explain course content, expectations, and requirements such as Vision and Mission, Program Educational Objectives, Student Outcomes, Quality Objectives and Quality Policy, and Grading System. 2. Recognize and differentiate between various programming languages, including their unique features and use cases.	Lecture/Discussion, Recitation Question and Answer Simulation Machine Problem	Computer Projector Board marker	Lab Exercises



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2	Tools for Developing Program Logic: Algorithm, Pseudocode Instructions, Flowchart	1. Learn about tools for developing program logic like algorithms, pseudocode, and flowchart. 2. Create an algorithm, pseudocode, and flowchart to represent the program's logic.	Simulation Machine Problem	Computer Projector Board marker	Laboratory Exercises
3	Variable Declaration, Data Types, Comments, Operators	1. Define what variables are and how they are used to store data in Python. 2. Convert data between different data types using type conversion functions (e.g., int(), float(), str()). 3. Recognize and use various Python operators, including arithmetic, comparison, logical, and assignment operators.	Simulation Machine Problem	Computer Projector Board marker	Laboratory Exercises
4					
5	Conditional Statements	1. Create if-else statements to execute different code blocks based on whether a condition is true or false. 2. Understand how to write and evaluate conditional expressions and conditions using comparison operators and logical operators.	Simulation Machine Problem	Computer Projector Board marker	Laboratory Exercises Online Certification (Solo Learn)
Prelim Examination					



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7	Looping Statements for loop, while loop, break and continue statement	1. Explain the fundamental concepts of loops in programming, including iteration, conditional repetition, and loop control. 2. Implement loop control mechanisms using break and continue statements to manipulate loop execution and exit conditions.	Simulation Machine Problem	Computer Projector Board marker	Laboratory Exercises
8	Functions Built-in functions, user- defined functions, Return Statement, Function Argument and Parameter, Passing Parameters, Default Parameters, Keyword Arguments	1. Create well-structured user-defined functions with appropriate names, parameters, and return types, adhering to naming conventions. 2. Recognize the distinction between built-in functions provided by the programming language and user-defined functions created by developers.	Simulation Machine Problem	Computer Projector Board marker	Laboratory Exercises
9					
10	List	1. Explain the fundamental concept of lists as a data structure in programming and their role in organizing and managing data 2. Access and retrieve individual elements from lists using indexing and slicing techniques	Simulation Machine Problem	Computer Projector Board marker	Laboratory Exercises



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11	Strings	1. Explain the fundamental concept of strings as a data type in programming and their role in representing and manipulating text data. 2. Utilize built-in string methods and functions. 3. Apply string formatting techniques to create well-structured and formatted output.	Simulation Machine Problem	Computer Projector Board marker	Laboratory Exercises Online Certification (Solo Learn)
Midterm Examination					
13	Dictionaries	1. Explain the fundamental concept of dictionaries as key-value data structures in programming and their role in organizing and managing data. 2. Access and retrieve values from dictionaries using keys, understanding the importance of key-value pairs for data retrieval. 3. Modify dictionary contents by adding, updating, or deleting key-value pairs, demonstrating proficiency in dictionary manipulation operations.	Simulation Machine Problem	Computer Projector Board marker	Laboratory Exercises
14	File Handling	1. Explain the fundamental concepts of file handling in programming, including	Simulation Machine Problem	Computer Projector Board marker	Laboratory Exercises



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		the role of files in data storage and processing. 2. Implement techniques to read data from files, including reading entire files, reading line by line, and reading specific portions of files. 3. Implement techniques to write data to files, including writing text, binary data, and structured data formats.			
15	Graphical User Interface, Databases	1. Develop GUI applications in Python using libraries such as Tkinter or PyQt, including creating windows, buttons, labels, and input fields.	Simulation Machine Problem	Computer Projector Board marker	Laboratory Exercises Software Development (Project)
16		2. Implement event-driven programming in Python GUIs to respond to user actions.			
17		3. Connect Python GUI applications to databases, perform CRUD (Create, Read, Update, Delete) operations, and execute SQL queries.			
Final Examination					



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TEXTBOOKS AND REFERENCE MATERIALS	Textbook: Pomperada J. R., et. al. (2019). Fundamentals of Python Programming. Mindshapes Co., Inc. Abante M. V., et. al. (2019). Python Programming. Unlimited Boks Library Services & Publishing Inc. Hudson D. (2020). Python Programming for Beginners. Larsen & Keller. Pomperada J. R., Soberano K. T. (2019). MySQL. Mindshapes Co., Inc. Abante M. V., et. al. (2019). Introduction to C. Unlimited Boks Library Services & Publishing Inc. Online References/Videos: "Sololearn," 2022. [Online]. Available: https://www.sololearn.com/learning/1073 . [Accessed 28 September 2022]. "w3schools," W3Schools, 2022. [Online]. Available: https://www.w3schools.com/python/ . [Accessed 28 September 2022]. "Tutorials Point," Tutorials Point India Private Limited, [Online]. Available: https://www.tutorialspoint.com/python3/index.htm. [Accessed 28 September 2022].																					
GRADING SYSTEM	1. The grading system is based 0, with 60% as the passing rate for courses/ units to be credited. 2. The periodical grade is rounded off to the nearest whole number before it is transmuted to grade point equivalent with 1.00 as the highest and 5.00 as the lowest. <table><tr><th>Grade Point Equivalent</th><th>Percentage Grade</th><th>Description</th></tr><tr><td>1.00</td><td>96 – 100</td><td>Excellent</td></tr><tr><td>1.25</td><td>92 – 95</td><td>Superior</td></tr><tr><td>1.50</td><td>88 – 91</td><td>Very Good</td></tr><tr><td>1.75</td><td>84 – 87</td><td>Highly Satisfactory</td></tr><tr><td>2.00</td><td>80 – 83</td><td>Good</td></tr><tr><td>2.25</td><td>75 – 79</td><td>Satisfactory</td></tr></table>	Grade Point Equivalent	Percentage Grade	Description	1.00	96 – 100	Excellent	1.25	92 – 95	Superior	1.50	88 – 91	Very Good	1.75	84 – 87	Highly Satisfactory	2.00	80 – 83	Good	2.25	75 – 79	Satisfactory
Grade Point Equivalent	Percentage Grade	Description																				
1.00	96 – 100	Excellent																				
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2.25	75 – 79	Satisfactory																				



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		2.50	70 – 74	Fairly Satisfactory
		2.75	65 – 69	Passing Minimum
		3.00	60 – 64	Passed
		5.00	0 – 59	Failed
		Other Remarks:		
		DRP		Dropped
		INC		Incomplete
		3. On Computation of Periodical and Final Grades for Collegiate		
		3.1. The final grades for laboratory courses are computed in the following manner:		
		FINAL GRADE = Prelim (30%) + Midterm (30%) + Finals (40%)		
COURSE REQUIREMENTS		3.2. The periodical and final rating for laboratory courses are computed in the following manner:		
		Laboratory: 40% (Laboratory Exercises) + 60% (Laboratory Exam)		
		Assessment Title	Assessment Category	Assessment Type
		Lab exercise 1	Case Study	Assignment
		Lab exercise 2	In-class assessment	Lab exercise
		Lab exercise 3	In-class assessment	Lab exercise
		Lab exercise 4	In-class assessment	Lab exercise
		Lab Exam	Term Exam	Prelim Exam
		PRELIM		
		Individual or group	Points	



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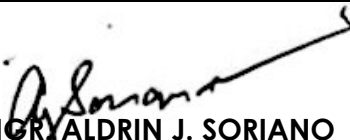
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	Lab exercise 5	In-class assessment	Lab exercise	Individual	
	Lab exercise 6	In-class assessment	Lab exercise	Individual	
	Lab exercise 7	In-class assessment	Lab exercise	Individual	
	Lab exercise 8	In-class assessment	Lab exercise	Individual	
	Lab Exam	Term Exam	Midterm Exam	Individual	
	MIDTERM				
	Lab exercise 5	In-class assessment	Lab exercise	Individual	
	Lab exercise 6	In-class assessment	Lab exercise	Individual	
	Lab exercise 7	In-class assessment	Lab exercise	Individual	
	Lab exercise 8	In-class assessment	Project	Individual	
	Lab Exam	Term Exam	Finals Exam	Individual	
	FINALS				
CLASS POLICIES	Class Attendance As per CHED policy on attendance, a student who misses a maximum of 20% of the total contact hours will be given a grade of failure due to absences. Academic Dishonesty – is a deliberate attempt to misrepresent, fabricate or otherwise tamper with data, information, records, or any other material that is relevant to the student's participation in any course, laboratory, or other academic exercise or function. These may include but may not be limited to: - Falsification, forgery, alteration, unauthorized possession, or misuse of school official documents, records, or identification, or knowingly furnishing false information to the university or to a university official. - Unauthorized accessing of electronic information of the university and securing, soliciting, and holding a copy of an examination or a portion thereof, that is yet to be administered. - Plagiarism or copyright infringement				



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	Submission of Course Requirements Course requirements must be submitted on time. Late submission means deduction.
EFFECTIVITY PERIOD	The course OBTL shall have three academic year effectivity. Necessary revision may still be done on a case-to-case basis, especially when new competency requirements need to be satisfied but cannot be immediately addressed by the current course OBTL.

Prepared by:	Reviewed and Recommended by:		Concurred by:	Approved by:
 ENGR. ALDRIN J. SORIANO Department Chair Date:	 ENGR. ALDRIN J. SORIANO Department Chair Date:	 PROF. MARIQUIT M. PEDRASA University Librarian Date:	DR. LUISITO L. LACATAN Dean, College of Engineering Date:	DR. FLORANTE A. MAGNAYE VP for Academic Affairs Date: